Part 1: Number Systems & Assembly Conversions

1. **How do you convert a decimal number to hexadecimal manually?** Divide the decimal number by 16 repeatedly and record the remainders. Read the remainders from the last one obtained to the first to get the hex value.
2. **How do you convert a binary number to decimal?** Multiply each bit (0 or 1) by its place value ($2^0, 2^1, 2^2$, etc.) starting from the right. Add all the results together.
3. **How do you convert a hexadecimal number to binary?** Replace each hexadecimal digit with its equivalent 4-bit binary group (e.g., A = 1010, F = 1111). Combine them to form the binary string.
4. **What is the difference between BCD and binary representation?** Binary represents the mathematical value of the whole number in base-2. BCD (Binary Coded Decimal) represents each individual decimal digit (0-9) separately using 4 bits.
5. **How can ASCII characters be converted to numeric values in assembly?** Subtract `30h` (decimal 48) from the ASCII character. For example, ASCII '5' (35h) minus 30h equals the integer 5.
6. **How do you store a decimal number in a register?** Use the `MOV` instruction followed by the register and value. For example: `MOV AX, 10` puts the decimal value 10 into the AX register.
7. **What instruction is used for arithmetic conversion in 8086?** `CBW` (Convert Byte to Word) is used to extend a signed byte into a word. Instructions like `AAA` and `DAA` are used to adjust formats after arithmetic.
8. **How do you convert a number from lower case to upper case using ASCII values?** Subtract `20h` (decimal 32) from the lowercase letter's ASCII value. For example, 'a' (61h) minus 20h becomes 'A' (41h).
9. **Explain signed vs unsigned conversion.** Unsigned conversion treats all bits as the magnitude (positive numbers only). Signed conversion checks the Most Significant Bit (MSB); if it is 1, the number is treated as negative.
10. **How do you represent negative numbers in 8086?** They are represented using **2's Complement**. To find this, you invert all bits of the positive number and add 1.
11. **How do you convert an ASCII digit to its integer value in 8086?** (Same as question 5) You subtract `30h` from the ASCII value.
12. **How is hexadecimal used in memory addressing?** Hex is used because it is shorter and easier to read than binary. Addresses are usually written as `Segment:Offset` in hex (e.g., `0700:1234`).

13. **What is the use of AAA instruction? AAA** (ASCII Adjust After Addition) corrects the result in the AL register after adding two unpacked BCD numbers so the result remains a valid decimal digit.
14. **Explain DAA instruction for BCD adjustment. DAA** (Decimal Adjust Accumulator) corrects the result of an addition to ensure the answer is a valid Packed BCD number (two decimal digits in one byte).
15. **How do you handle overflow during conversion?** Check the **Overflow Flag (OF)** in the flags register. If OF=1, the result is too big to fit in the destination register.